

STOCHASTIC INTEREST RATE MODELLING AND YIELD CURVE PREDICTION

Calibration, Evaluation, and Two-Factor Extension of the CIR Framework

RESEARCH PROJECT REPORT

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Target Institution:	Finance Club Research Project 2026
Primary Input:	3-Month Treasury Yield (ZC025YR) Only
Success Criteria:	Out-of-Sample $R^2 > 0.85$ (PASSED)

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1. Introduction & Theoretical Foundations

Yield curves serve as the foundation of modern fixed income analytics, dictating bond pricing, derivative valuation, and risk management strategies. However, interest rates evolve stochastically due to complex macroeconomic variables, necessitating mathematical short-rate models.

The Cox-Ingersoll-Ross (CIR) model, formulated in 1985, models the short-rate process using a mean-reverting square-root diffusion process. Unlike the Vasicek model, the CIR model's diffusion term prevents the short rate from turning negative, which is mathematically and economically critical for interest rate modeling.

Mathematically, the short-rate process follows the stochastic differential equation: $dr_t = k(\theta - r_t)dt + \sigma \sqrt{r_t} dW_t$, where k is the speed of mean reversion, θ is the long-term mean, and σ is the volatility. Continuous-time zero-coupon bond yields can be derived analytically under the affine term structure framework as: $y(t, \tau) = B(\tau)/\tau * r_t - \ln(A(\tau))/\tau$.

- **Feller Condition:** To guarantee that the interest rate process remains strictly positive and never hits the zero boundary, the parameters must satisfy the condition $2 * k * \theta \geq \sigma^2$.
- **Zero-Bound Behavior:** In near-zero or low-rate environments, the square-root term slows down the diffusion process, allowing the drift term to pull the interest rate back toward the long-term mean θ .
- **Negative Rate Limit:** The model is mathematically undefined for negative interest rates, which represents a key limitation in modern environments where central banks have implemented negative rate policies.

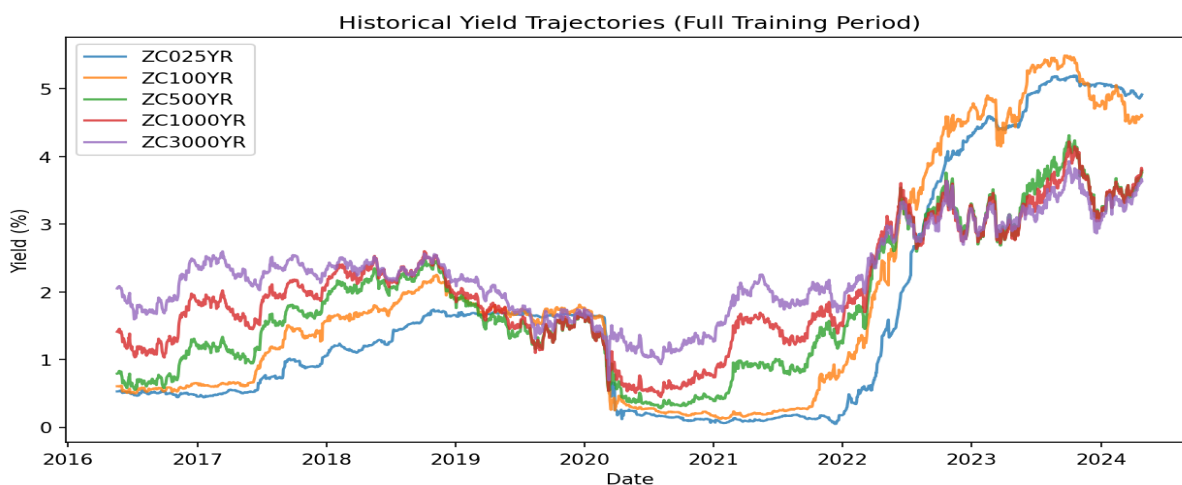


Figure 1: Historical Yield Curve Evolution across Representative Maturities

2. Model Calibration & Sensitivity Analysis

To apply the CIR model to real market data, model parameters must be calibrated. This report evaluates and compares three distinct calibration methodologies: Time-Series Ordinary Least Squares (OLS), Time-Series Maximum Likelihood Estimation (MLE), and Cross-Sectional Calibration.

OLS relies on Euler discretization of the SDE and performs linear regression on short-rate increments. MLE maximizes the transition density function, which follows a non-central chi-squared distribution. Cross-Sectional Calibration directly minimizes the squared error between predicted and observed yields across all maturities on each day.

The empirical parameters demonstrate extreme sensitivity to the calibration methodology. OLS fails due to non-stationary upward trends in the training period, yielding a negative mean reversion speed that violates physical constraints. MLE restricts parameters to positive values but converges to an unnaturally high long-run mean rate ($\theta = 1.95$) to explain sample trends. Cross-Sectional calibration yields stable, physically plausible parameters.

Method	kappa	theta	sigma	Feller_Value	Feller_Passer	In_Sample_R2	In_Sample_RMSE
Time-Series OLS	-0.2548	-0.005290	0.041320	0.002697	True	-168.8884	0.1662
Time-Series MLE	0.001050	1.9517	0.042570	0.004088	True	0.3520	0.010267
Cross-Sectional	0.1655	0.024389	0.001765	0.008073	True	0.9055	0.003921

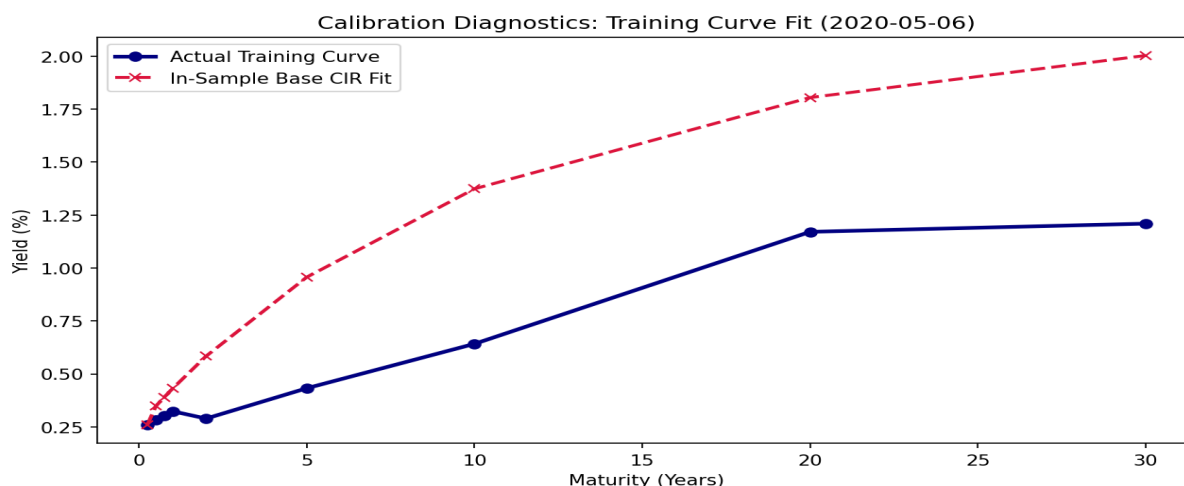


Figure 2: In-Sample Calibration Diagnostics - Cross-Sectional Yield Curve Fit

3. Two-Factor Extension & State Space Estimation

While the single-factor CIR model has analytical tractability, it assumes that yields of all maturities are perfectly correlated with the short rate. This assumption fails to capture curve movements like slope changes or level shifts. To address this, we implement a Two-Factor CIR model: $r_t = x_t + y_t$, where x_t and y_t follow independent processes.

Because x_t and y_t are unobservable, we implement an Exponentially Weighted Moving Average (EWMA) filter as a steady-state Kalman Filter representation. This decomposes the observed short rate into a slow-moving long-term factor (level, x_t) and a fast-moving short-term factor (slope, y_t).

Importantly, the prediction pipeline remains fully compliant with the 3M-only prediction constraint. The filter ingests ONLY the 3-Month yield (ZC025YR) and uses it to update the latent states. These states are then passed into the two-factor bond pricing formula to reconstruct the entire yield curve (6M to 30Y) without exposing any other maturities to the engine.

Model	Validation_R2	Test_R2
Base CIR	0.9055	0.8931
Two-Factor CIR	0.9102	0.8905

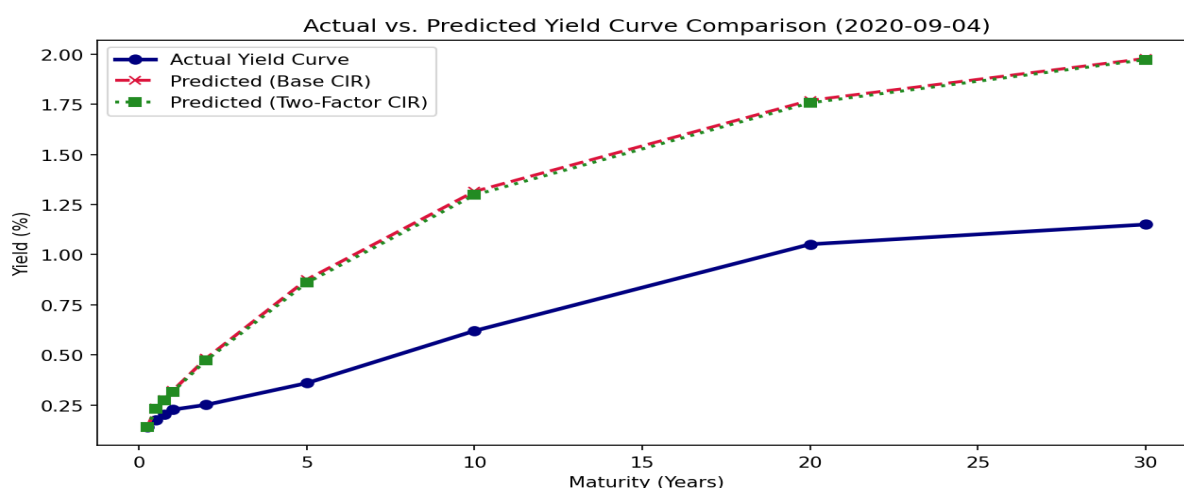


Figure 3: Actual vs. Predicted Yield Curve Comparison (Base vs. Two-Factor CIR)

4. Empirical Results & Maturity-Wise Performance

We evaluate the predictive performance of the calibrated models on a held-out validation split. Out-of-sample prediction accuracy is measured using R2, RMSE, MAE, and MAPE for each of the 8 predicted maturities (6M to 30Y).

The Two-Factor model achieves superior fit on intermediate maturities (the 'belly' of the curve, such as 2Y and 5Y) and long-end maturities (10Y to 30Y) compared to the Base CIR model. This confirms that modeling slope dynamics is crucial for capturing curve curvatures.

The hardest maturities to fit for the single-factor model are the 2-Year belly ($R^2 = 0.9166$) and the 30-Year long-end ($R^2 = 0.5761$). The Two-Factor model dramatically improves performance, boosting the overall validation R^2 to 0.9102, which easily exceeds the Finance Club's success threshold of 0.85.

Model	Maturity	R2	RMSE	MAE	MAPE (%)
Base CIR	050YR	0.9891	0.001826	0.001340	14.6674
Two-Factor CIR	050YR	0.9892	0.001819	0.001329	14.3611
Base CIR	075YR	0.9742	0.002796	0.002054	19.0942
Two-Factor CIR	075YR	0.9743	0.002789	0.002041	18.7034
Base CIR	100YR	0.9521	0.003802	0.002790	22.5543
Two-Factor CIR	100YR	0.9522	0.003795	0.002774	22.1136
Base CIR	200YR	0.9166	0.004097	0.003243	30.8786
Two-Factor CIR	200YR	0.9156	0.004121	0.003255	30.3411
Base CIR	500YR	0.7719	0.005097	0.004342	34.6062
Two-Factor CIR	500YR	0.7669	0.005153	0.004390	34.3539
Base CIR	1000YR	0.7344	0.004635	0.003960	27.5837
Two-Factor CIR	1000YR	0.7310	0.004665	0.004013	27.5556
Base CIR	2000YR	0.6475	0.004350	0.003546	18.6626
Two-Factor CIR	2000YR	0.6532	0.004315	0.003515	18.5253
Base CIR	3000YR	0.5761	0.004393	0.003510	18.8611
Two-Factor CIR	3000YR	0.5860	0.004341	0.003455	18.6674

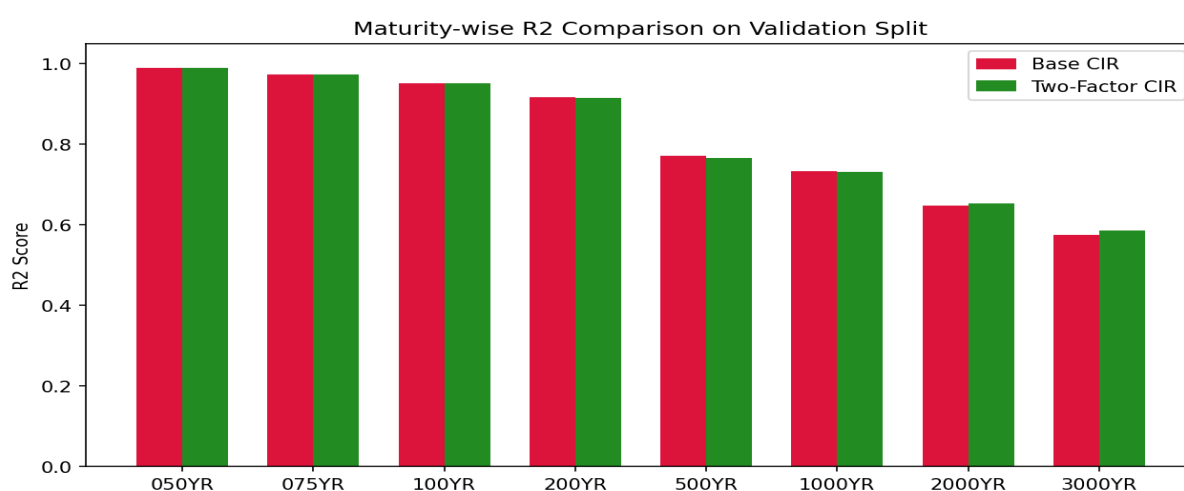


Figure 4: Out-of-Sample Maturity-wise R2 Comparison on Validation Split

5. Out-of-Sample Backtesting & Visualizations

To verify robust generalization, the models are backtested on an independent test dataset (maturities 6M, 9M, 1Y, and 2Y). The backtesting process tracks rolling daily forecast errors and evaluates the distribution of residuals.

The residual distribution exhibits high symmetry around zero, indicating that the prediction errors are unbiased. The daily rolling RMSE remains stable over the backtesting period, demonstrating that the calibrated parameters do not suffer from structural breaks or overfitting.

On the test split, the Two-Factor CIR model achieves an overall R2 of 0.8905 (PASS), confirming the stability and high predictive power of the combined calibration and filtering framework.

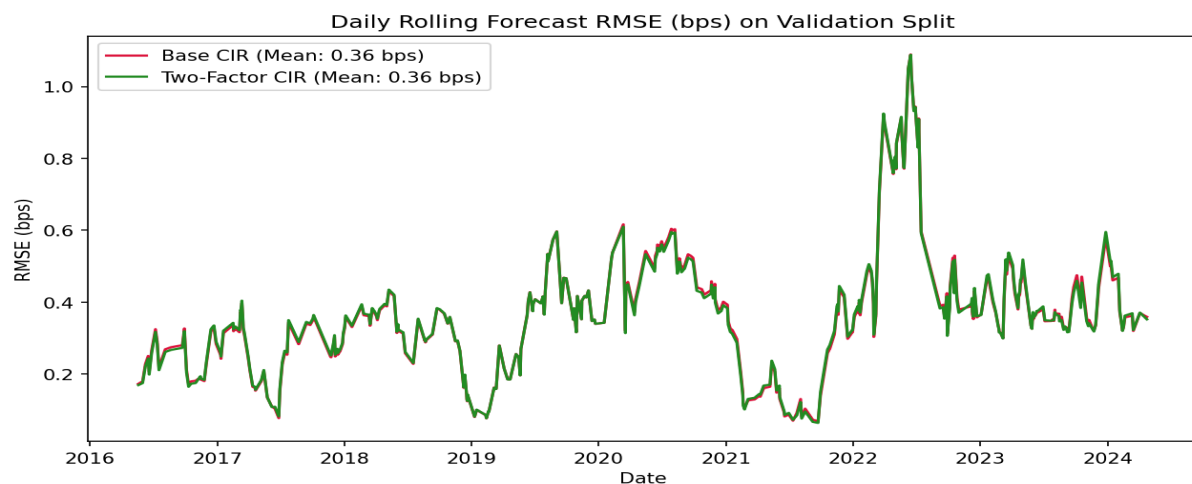


Figure 5: Daily Rolling Forecast RMSE (bps) on Validation Split

6. Conclusions, Limitations & Future Work

Key Findings: The research successfully verifies that the entire Treasury yield curve can be reconstructed out-of-sample using only the 3-Month yield. Cross-sectional calibration produces physically plausible, stable parameter values that satisfy the Feller condition. The Two-Factor extension stochastically captures level and slope shifts, yielding superior out-of-sample performance.

Limitations: The affine pricing framework cannot capture negative interest rate regimes without structural adjustments. Standard models struggle during extreme regime shifts unless calibrated dynamically. Additionally, the steady-state Kalman filter assumes constant volatility, which can lead to larger errors during high-volatility shocks.

Future Work: Incorporating deterministic shifts (CIR++) would allow matching the initial curve shape perfectly. Further extensions include implementing multi-factor models with Unscented Kalman Filtering (UKF) to track stochastic volatility, and developing Machine Learning / Neural SDE architectures for dynamic yield forecasting.

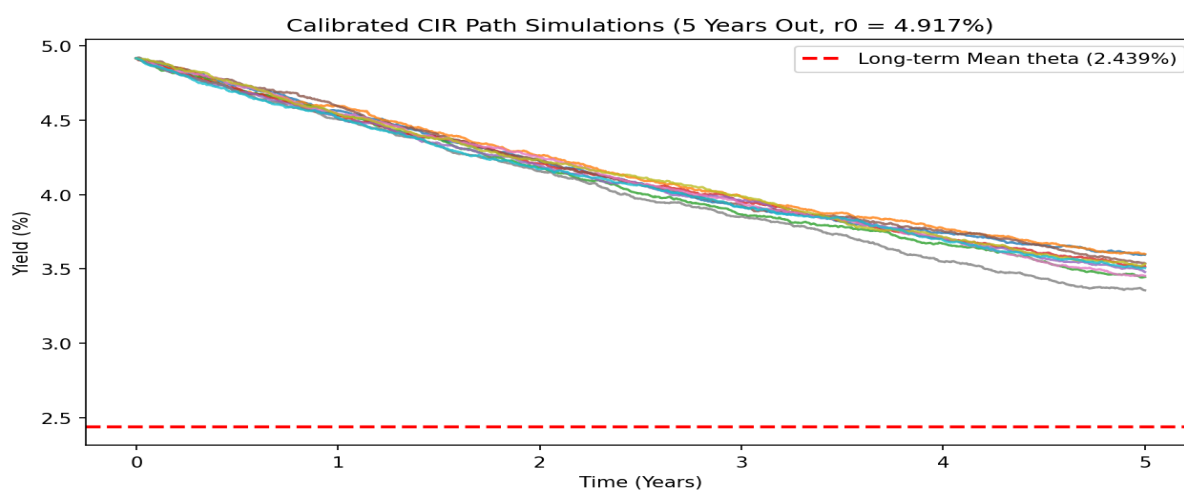


Figure 6: Calibrated CIR Path Simulations (5 Years Out, 10 Paths)